

Role Enactment of Multi-User LLM-Based Conversational Agents in Group-Based Knowledge Work: A Pilot Study of Collaborative UX Design

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Abstract

LLM-based conversational agents are increasingly being integrated into knowledge work; however, most existing systems are designed primarily for single-user interaction. Our research investigates an emerging category of systems - Multi-User LLM-based Conversational Agents (MULCAs)—and examines their potential to function as collaborators in group-based knowledge tasks. As a case study, we focus on collaborative UX design and report findings from an exploratory study involving 24 MSc students. Participants worked in small groups and interacted with a MULCA during the initial tasks of the UX design process. We collected data on participants' initial expectations regarding MULCA support, their patterns of use throughout the UX design process, the agent's observable behaviors, and their perceived utility of the system. Our findings highlight the diverse roles MULCAs can assume in collaborative UX design and provide a deeper understanding of both the opportunities and limitations of such agents in supporting group-based knowledge work.

CCS Concepts

• Information systems → Collaborative and social computing systems and tools.

Keywords

Large Language Models, Multi-User Conversational Agents, Human-Agent Collaboration

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1 Introduction and Background

As Large Language Model (LLM)-based systems take on larger and more autonomous roles within knowledge work, recent technological advances open up possibilities for moving them from individual tools to teammates in group-based knowledge-intensive activities that involve discussion and negotiation among multiple participants. Effective partnership in these activities requires Multi-User LLM-based Conversational Agents (MULCAs), i.e., intelligent systems that can manage dialogues with and among multiple interlocutors, simultaneously, and in a shared conversation space. MULCAs would represent a fundamental shift in technology-supported collaborative work [16], engaging participants more naturally and adaptively than structured interfaces [7, 12]. MULCAs are particularly complex agents since they must reason about what to say, when to say it, and who to say it to [11], monitoring the dialogue state and participant contributions to ensure contextually relevant responses over time [7, 11], and their design and development require a clear understanding of the roles they can play across different knowledge work contexts. Among prior research exploring this issue, [17] reports an empirical investigation of human expectations on AI-based systems acting as coequal partners in collaboration scenarios. Based on the results of a vast quantitative survey and nine expert interviews, this research identified four desired team roles for AI-based teammates - coordinator, creator, perfectionist and doer, which aligned with Belbin's framework [1]. Some studies have explored how MULCAs can fulfil a specific role in a specific collaborative task. As *brainstormers*, LLMs stimulate [3] and generate ideas [10], or refine existing ones [18]. As *knowledge provider*, they can retrieve information [4], and organise diverse information and perspectives [14]. As *collaboration facilitators*, they structure discussions and mediate conflicts [9]. As *Task Supporters*, they can help to plan and decompose tasks [6, 15] and provide support during task execution [2]. In the field of multi-agents technology, some research has explored the collaboration of multiple LLMs in multi-user activities. This is achieved by assigning a specific role to each LLM collaborating with humans, or embodying specific character traits (personas) in each LLM [8]. In some cases, complex activities are decomposed in simpler tasks and specialised agents are assigned to each subtask [4, 13], mutually collaborating on planning task assignments [5].

Despite increasing interest in MULCAs, our understanding remains limited regarding how a multi-user conversational agent *based on a single LLM* can navigate *multiple roles* to support collaborative *multi-task* processes. To help address this gap, we conducted an observational study involving 24 students enrolled in a graduate-level HCI course, who used a MULCA in small groups during early-stage UX design. The MULCA was based on ChatGPT-4o using an intentionally minimalist prompt-only approach rather than a heavy custom architecture.

The study explores two main research questions: **(RQ1)** *What roles do team members expect from a MULCA during early-stage UX design?* **(RQ2)** *What roles can a lightweight MULCA actually perform most effectively when collaborating with humans during this process?*

Our findings explore perceptions of which MULCA roles are perceived as most valuable for collaborative UX design, and how well these roles are supported by a single, prompt-engineered, off-the-shelf LLM. From a broader perspective, our work highlights both the opportunities and the limitations associated with the transition of this class of intelligent agents from individual tools to teammates for multi-user, multi-task cooperative processes.

2 Procedure

The 24 participants (13 men, 11 women, aged 22-25) were enrolled in a graduate-level project-based HCI course, which required students to work in small self-formed groups (three members) to design and prototype an interactive system in a domain of their choice. Participants had a homogeneous level of basic knowledge of user-centered design. Participation in the study was voluntary, and the protocol was approved by the university's ethics committee. Participants were asked to collaborate on the initial tasks of the UX design process over the course of one month, using an online platform (Discord) integrated with MULCA. They were required to use the system to discuss, ideate, and make decisions regarding: elicitation of profiles and needs of the various stakeholders of their interactive technological artifact; definition of the context of use for the system-to-be; definition of design constraints, user scenarios, and key UX features. On the platform, students could communicate with each other using either voice or text, but only textual messages were shared with the MULCA. Conversations with the MULCA were visible to all team members and were exclusively text-based. To ensure consistent data collection, participants were asked to use only this collaboration mode during the various tasks, and were instructed not to use any external LLM-based tools during their work. Prior to the study, participants were briefly introduced to four functional roles identified in the current literature on human-LLM cooperation that MULCAs could play in collaborative UX design: *brainstormer*, *knowledge provider*, *collaboration facilitator*, and *task supporter* [3, 4, 9, 15]. This aimed to guide students' use of the MULCA and establish a shared vocabulary for reflecting on its contributions. At the start of the study, participants completed a *pre-questionnaire* to assess familiarity with LLMs and to capture their expectations regarding the MULCA's role(s) and contribution during early UX design process. During the study, participants were required to keep track of their collaborative activity and to complete

an *individual diary* for each design session, reporting the tasks performed and providing observations on how they used the MULCA and how the agent behaved. We also collected the *chat logs* of all textual conversations, both among students and between students and the MULCA. After completing all tasks, participants filled out a final questionnaire. Through closed- and open-ended questions, the *post-study questionnaire* examined MULCA's perceived utility across the various tasks and how participants experienced the MULCA's role enactment during the collaborative UX design process. The MULCA was implemented as a ChatGPT-4o textual bot embedded in Discord using a Python-based integration with the OpenAI API. This is a proprietary model that is not open with regards to its training data, training process, or model weights. This limits the reproducibility of our study, but reflects a deliberate choice to use the type of model that students and UX practitioners are most likely to encounter in practice - a trade-off that aligns with our research goals. The agent was system-prompted with instructions specifying the design tasks it was expected to support, as well as its goals as a virtual tutor for the co-design team (e.g., facilitating collaborative UX design by encouraging reflection, asking clarifying questions, providing structure to support higher-order thinking, and supporting design decision-making by guiding rather than directly solving tasks). The agent was also instructed to provide responses limited to 450 words in order to comply with Discord's message limit.

3 Results

For lack of space, this section focuses on the comparison between the participants' expected roles of the MULCA (reported in the pre-questionnaires) and the roles perceived during the study (emerging from the post-questionnaires). Through thematic analysis of *pre-study questionnaires*, we identified five expected roles encompassing 17 distinct behavioral modes (Table 1). While participants were introduced to four literature-based roles, inductive coding revealed **Reviewer** as a fifth role, reflecting LLMs' ability provide feedback and refine the user's input. An independent post-study analysis identified the same five roles and 14 of the same modes, and revealed which roles were successfully enacted.

Post-study questionnaires highlight that four expected modes were not observed in practice. The initial implementation of MULCA was not sufficiently able to *structure discussions* or *resolve conflict*, because "*support was reactive and on demand, rather than proactive or continuous*" [P11]. Participants likely did not experience the *analyze information* function because they did not perform intensive information processing tasks. Participants also did not report that MULCA *checked the correctness* of their ideas, likely because MULCA automatically offered suggestions to *refine ideas* instead. The distinction between the **Brainstormer's** *expand ideas* mode and the **Reviewer's** *refine ideas* mode is in divergent versus convergent thinking.

Both the pre- and post-study analysis indicated we should expand the **Knowledge Provider** role into the **Knowledge Manager** role, encompassing not only the provision of information, but also its processing functions (organizing, summarizing, tracking).

Additional themes were found during the post-study analysis for the **Brainstormer** role, as it proved particularly strong at *generating examples and scenarios*, and in *helping design interface elements*.

Table 1: The roles a MULCA can play based on user expectations and experiences, and which were successfully enacted. Based on independent thematic analyses of pre- and poststudy questionnaires of group projects with an agent.

Role	Mode	Explanation	Expect.	Exper.	Success.
Brainstormer	Generate ideas	Introduce a novel idea	✓	✓	✓
	Stimulate ideas	Stimulate user towards new ideas	✓	✓	✓
	Expand ideas	Creatively add to proposed ideas	✓	✓	✓
Knowledge Manager	Provide information	Provide new information	✓	✓	✓
	Organize information	Provide a structure to information	✓	✓	✓
	Summarise information	Summarise key points	✓	✓	✓
	Keep track of information	Monitor user contributions	✓	✓	×
	Analyze information	Compare and analyze ideas	✓	×	×
Task Supporter	Structure workplan	Provide a path forward	✓	✓	×
	Maintain focus	Focus conversation on goal	✓	✓	×
	Provide guidance	Suggest a step forward	✓	✓	✓
Communication Facilitator	Structure discussion	Leading the discussion	✓	×	×
	Balance participation	Encourage equal speaking time	✓	✓	✓
	Resolve conflict	Mediate communication issues	✓	×	×
	Check correctness	Point out unmentioned problems	✓	×	×
Reviewer	Evaluate	Reflect if idea is feasible, goal-aligned and satisfies practical constraints	✓	✓	✓
	Refine ideas	Suggest improvements, concretise ideas	✓	✓	✓

These represent domain-specific use-cases that are supported by the three functional modes of the brainstormer rather than distinct roles. Similarly, the theme *support exploration and divergent thinking* is a result of the brainstormer’s roles, rather than a distinct role.

MULCA performance varied across roles. **Brainstormer**, **Reviewer**, and most **Knowledge Manager** functions were perceived as useful, though not without limitations (e.g. “its input was sometimes generic or broader than needed” [P9]. However, MULCA was not proactive enough to perform the **Task Supporter** and **Communication facilitator** roles effectively.

4 Discussion and Conclusions

The roles we identified in our study align with the four primary roles identified in Siemon’s factor analysis [17] of what humans expect from AI teammates. Our key extension is the *knowledge provider* role, reflecting LLMs’ unique retrieval capabilities. While the MULCA performed most expected roles satisfactorily, the findings of our study reveal some mismatches between users’ expectations and the MULCA’s actual behavior. Users expected MULCA to *structure discussions* and *analyze information*, yet they did not report using the system in these ways. Additionally, users indicated that MULCA underperformed in *keeping track of information*, *structuring the workplan*, and *maintaining focus*. We might attribute this gap to both the nature of tasks performed and MULCA’s functional limitations. The collaborative tasks did not require extensive information analysis, with basic processing adequately handled by the idea refiner, information organizers, and information summarizer sub-roles. The remaining facilitation roles provided difficult to full-fill because the MULCA lacked sufficient proactivity to effectively guide the collaboration process.

To our knowledge, this is the first empirical study to investigate both the roles users need from a conversational virtual teammate during complex knowledge work and the perceived performance

—during real collaborative tasks — of a multi-user bot based on a single, lightweight LLM embodying multiple roles. The results of the study indicate that our MULCA demonstrated utility across several roles and highlight opportunities for transitioning LLM-based systems from individual tools to teammates in multi-user, multi-task collaborative processes. Our findings also suggest that MULCAs hold significant potential in educational contexts, particularly as virtual assistants supporting cooperative learning. At the same time, our study reveals important limitations. These could be addressed by enabling *more proactive behavior* and extending the MULCA *multimodally* to support participation in—or at least listening to—*spoken* discussions. Such improvements would likely require a more sophisticated architecture than our prompted-only, off-the-shelf LLM approach can provide.

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